

HumanEval Training Dynamics Analysis

RankAlign vs. TC methods — why Upper and Multi differ

Automated Analysis Report (in progress)

June 19, 2026

Abstract

This report mirrors the earlier RankAlign training-dynamics analysis (`analysis/report.tex`, membership/IFEval) but for the **HumanEval** task, across **gemma-4-31b** and **qwen-3.5-9b**, comparing the two HumanEval datasets **v2.1correct-upper** and **v2.1correct-multi**. It targets three questions: **(Q1)** why does RankAlign (s2) beat our TC methods (s4 self-TC, s7 neg-TC) on *multi*, while our methods win on *upper* (gemma-4)? **(Q2)** is typicality-correction (TC) training helping? **(Q3)** are the negative results on the test tables an *overfitting* effect, or are they already present on the train split? All generator metrics are typicality-corrected (tc) gen ROC AUC unless noted; each method is shown in its natural eval mode (RankAlign/s3 → Neg base; s4 → PMI base; s7 → Neg base), matching the convention in `docs/rerun_only_tables.tex`.

Status (2026-06-19): the TEST-set comparison (all settings, both models, both datasets) is complete. The train-set dynamics currently exist only for **gemma-4 s2/s4** (GROUP B). Train-set evals for **gemma s3/s7** and **all qwen settings** are running on mll (38-job `trs2` batch); sections depending on them are marked

TODO:

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1 Q1: The Upper-vs-Multi Effect (TEST set)

Table 1 reports tc gen ROC AUC ($\times 100$) on the held-out test split for every setting, each in its natural eval mode. The key pattern your question rests on:

- **gemma-4 / upper**: our methods *win* — FLORA-Neg (s7) = 94.8 and FLORA-PMI (s4) = 90.7 both beat RankAlign (s2) = 88.2.
- **gemma-4 / multi**: RankAlign *wins* — s2 = 94.2 vs s7 = 92.0, s4 = 90.1.
- **qwen**: RankAlign wins on *both* datasets (multi especially, 96.5). The “ours-wins-on-upper” effect is **gemma-4-specific**.

Table 1: HumanEval TEST tc gen ROC AUC ($\times 100$). Each method in its natural eval mode (Base: self; SFT/s4/s13: PMI base; RankAlign/New+fsx/s7: Neg base). gemma-4 has no base eval.

Model	Data	Base	SFT	RankAlign	New+fsx	FLORA-PMI (self-TC)	FLORA-Neg (neg-TC)
gemma-4-31b	upper	—	82.9	88.2	87.9	90.7	94.8
gemma-4-31b	multi	—	87.9	94.2	94.0	90.1	92.0
qwen-3.5-9b	upper	69.1	87.9	90.2	92.6	83.2	88.9
qwen-3.5-9b	multi	69.0	89.4	96.5	96.7	89.8	87.9

TODO: Add the train-set counterpart of Table 1 once the **trs2** evals land (gemma s3/s7 + all qwen). Then

2 Q3: Train vs. Test — Is It Overfitting?

This is the central question. For the settings we already have on both splits (**gemma-4 s2 RankAlign** and **s4 FLORA-PMI**, epoch 2), Table 2 compares train-split and test-split tc gen ROC.

Table 2: gemma-4 train vs. test tc gen ROC ($\times 100$), epoch 2, each method in its natural eval mode. Train = --train split, N=50 stratified candidates/problem. Δ = train – test (large positive $\Rightarrow$ overfitting).

Data	Setting	Train	Test	Δ (train–test)
upper	RankAlign	83.4	88.2	-4.8
upper	FLORA-PMI (self-TC)	90.9	90.7	0.2
multi	RankAlign	94.6	94.2	0.4
multi	FLORA-PMI (self-TC)	99.0	90.1	8.9

Early finding (gemma-4, s2 vs s4). On **multi**, FLORA-PMI (s4) shows a large train $\rightarrow$ test drop (99.0 $\rightarrow$ 90.1, Δ = +8.9) while RankAlign generalizes almost perfectly (94.6 $\rightarrow$ 94.2, Δ = +0.4). On **upper**, s4 does *not* overfit (90.9 $\rightarrow$ 90.7, Δ = +0.2). This is consistent with the hypothesis that **our TC method overfits on multi** — it is best on the training candidates but loses the advantage on held-out problems, which is exactly where RankAlign overtakes it at test time. RankAlign’s small train $\rightarrow$ test gap suggests it learns a flatter, more transferable ranking on this task.

TODO: This only covers s2 vs s4. The method that *wins* upper is **s7 (neg-TC)**, whose train-set eval is in t

TODO: Add the qwen train-vs-test rows (s2/s3/s4/s7) once **trs2** lands — needed to confirm whether qwen

3 Q2: Is TC Training Helping? (Focused TC comparison)

The clean test of TC is s3 (New+fsx, *no* TC) vs s4 (adds self-TC) and s7 (adds neg-TC), evaluated in the matching TC mode.

TODO: Fill the test-side TC-delta table (s3→s4 in self mode, s3→s7 in neg mode) for both models and datasets.

4 Q2 (cont.): Training Diagnostics — WandB

WandB logging was *online* for all HumanEval runs (project `rankalign`; run names `g4-{cu,cm}-s{S}` and `q35-{cu,cm}-s{S}`).

TODO: Pull and plot the loss curves (SFT vs RankAlign vs Ours; TC comparison) for gemma-4 and qwen-72b.

5 Diagnostics: Concordance / Correlation / Scatter

TODO: Compute generator–validator concordance, Spearman, and Pearson (epoch 2) and the gen-vs-val scatter.

6 Conclusions (preliminary)

1. **The upper-vs-multi flip is real and gemma-4-specific:** our TC methods (s7, s4) beat RankAlign on gemma-4 *upper* but lose on *multi*; on qwen RankAlign wins both.
2. **Preliminary overfitting signal:** gemma-4 FLORA-PMI overfits on multi ($\Delta = +8.9$ train–test) but not on upper ($\Delta = +0.2$), while RankAlign generalizes on both. This is consistent with the multi loss being an overfitting effect.

TODO: Finalize conclusions once s7/s3/qwen train-set evals + WandB + concordance land. Open questions.

Reproducibility. Tables built by `analysis/scripts/build_he_report_tables.py` from `docs/he_harvest_2026-06-19` (test) and `docs/he_trainset_2026-06-19/humaneval_TRAIN_metrics_aggregated.csv` (train, gemma s2/s4). trs2 train-set batch: `scripts/ml1_he_trainset_eval_v2.sbatch + launch_he_trainset_extra.py`. model origins in `docs/he_trainset_eval_model_origins.md`.